Lecture 3 - Evaluation

ECE 447: Data Analysis and Machine Learning for Engineers

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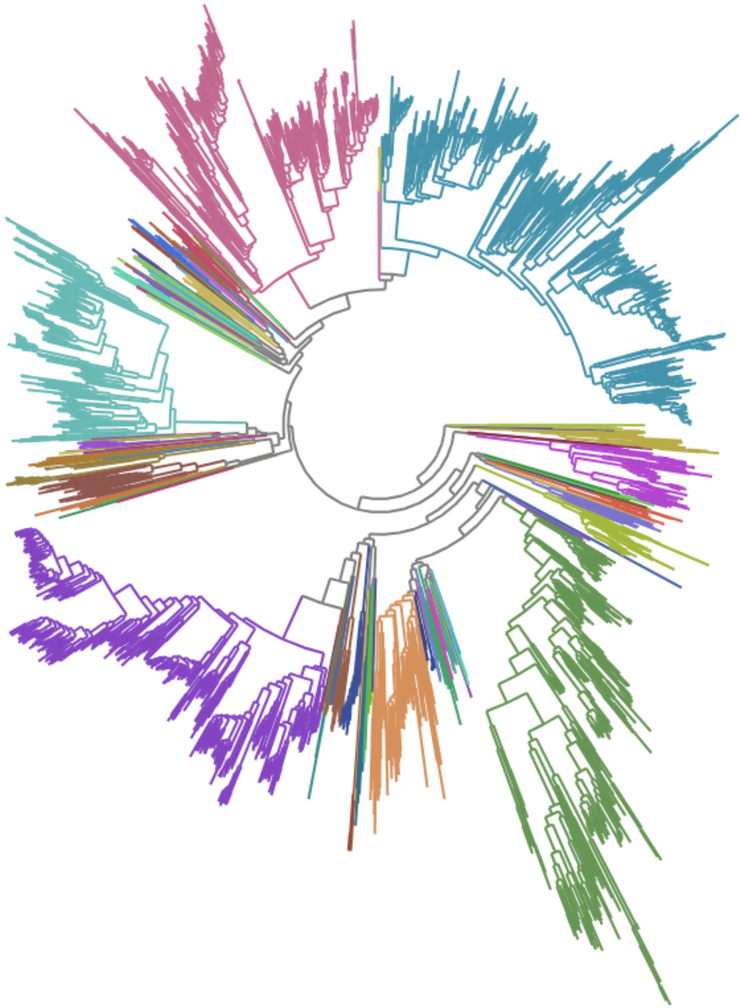
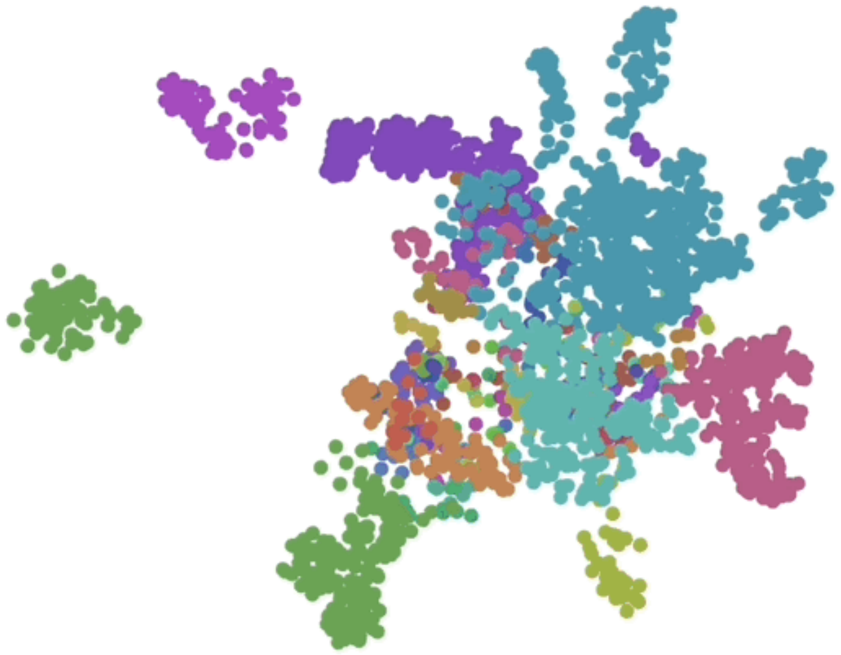
Mechanistic Interpretability of Large Models

This slide is informational, excluded from assessment.





Understanding model ontologies: how does EVO 2 represent the tree of life?

https://www.goodfire.ai/research/phylogeny-manifold#

This Lecture Covers ...

- Standard Approach
  - Testing/training, and confusion matrix.
- Evaluation Experiment
  - Sampling, k-fold cross validation, leave-one-out, and bootstrapping.
- Performance Measure
  - Categorical target.

— *ML for Predictive Data Analytics*

Create A Sample with Equally Represented Categorical Features

from last lecture

Under-sampling

Divide data into groups: no change for <u>smallest</u> group, and other groups are randomly sampled to the size of the smallest group.

Over-sampling

Divide data into groups: no change for <u>largest</u> group, and other groups randomly sampled with replacement to the size of the largest group.

Which one is preferred?

Over-sampling. You don't wanna remove valuable data.

Goals of Evaluation

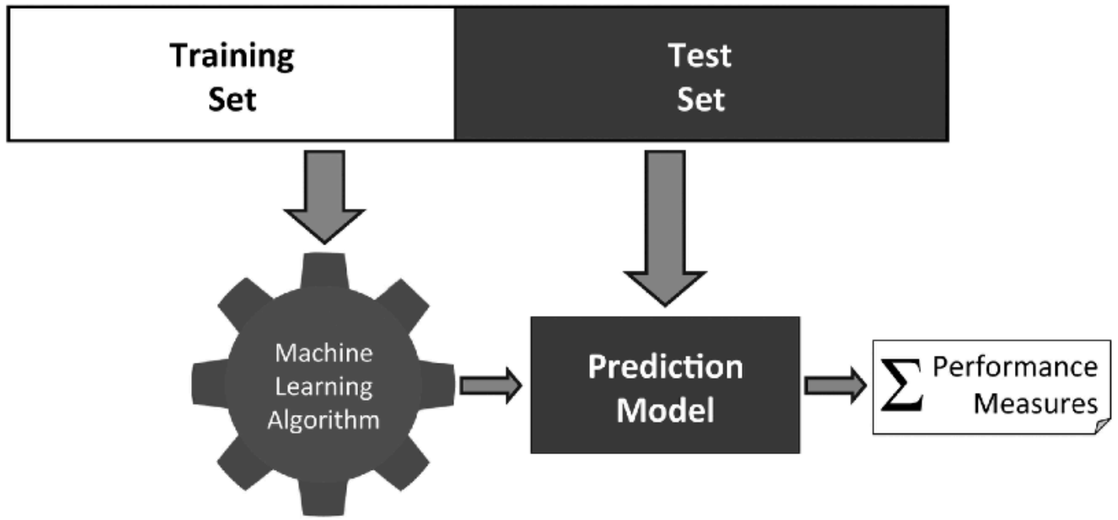
Can the generated model do the job it has been built for?

The purpose of **evaluation**:

- Which model is the best?
- How will the model perform when it is deployed?
- How to convince the "customer" the model meets their needs?

Evaluation - Standard Approach

Generalization *beyond the training data.*



Evaluate your model on a **test** set using a **performance measure**.

figure from ML for Predictive Data Analytics textbook.

Lets Look Into Two Key Aspects of Evaluation Together

- The **test set** to evaluate your model on.

- The **performance metric** used evaluate your model with.

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Evaluation Experiment: How to Choose the Test Set?

Hold-out Sampling

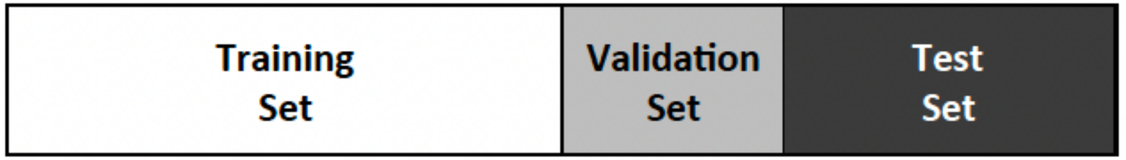
"Standard" process of model evaluation using ***test dataset*** *created by random sampling of the whole data.*

<u>Extension</u>:

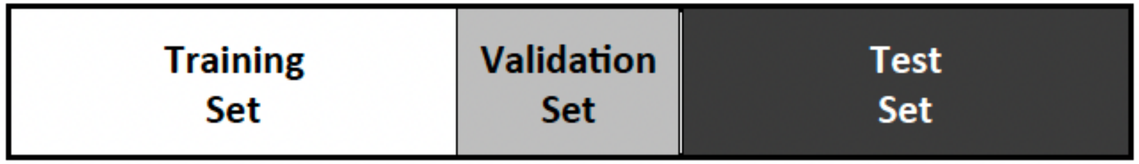
Put aside a random sampled dataset for ***validation****, for tuning particular aspects of the model (during your process of training).*

Common Ratio of Training/Validation/Test

Hold-out Sampling with Validation Set



(a) A 50:20:30 split



(b) A 40:20:40 split

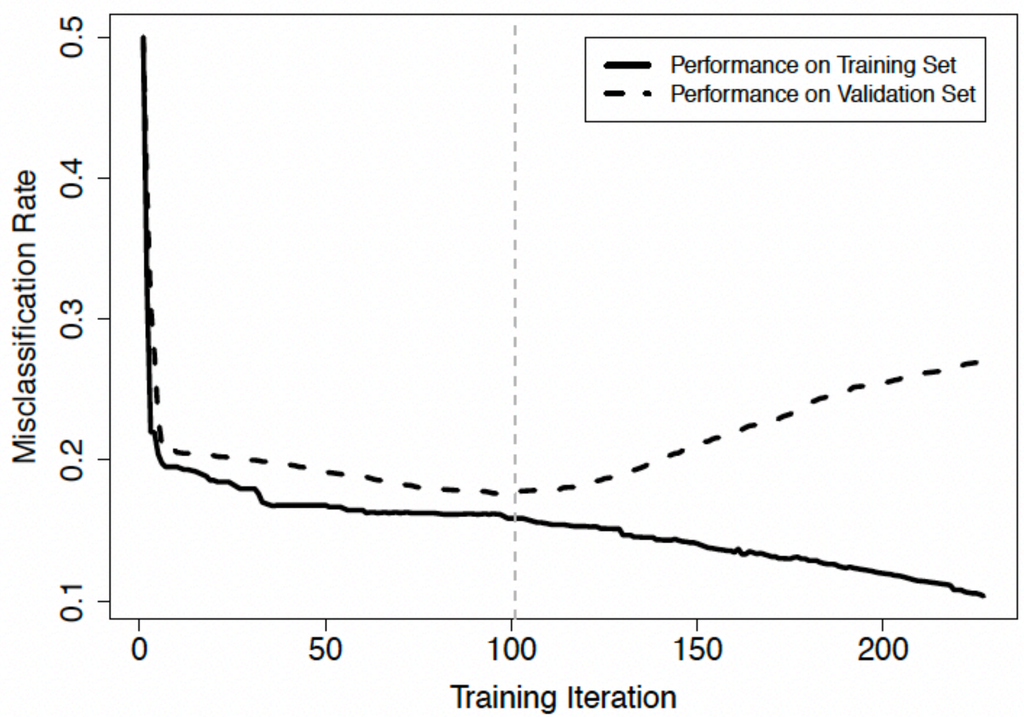


figure from ML for Predictive Data Analytics textbook.

Notes on Hold-out Sampling

1. Requires large data available to make suitable large training, test, and if required validation sets.

2. Measured performance could be misleading; we can have **lucky split** that has difficult cases in training, and simple ones in testing.

What to do to resolve this concern?
(If the cost of training is not much)

K-Fold Cross Validation (I)

Divide available data into **k equal-sized folds** (parts), and perform **k separate evaluations.**

1st evaluation: 1st fold as testing, $k-1$ next folds as training.
2nd evaluation: 2nd fold as testing, the rest $k-1$ as training.
...
k^{th} evaluation: k^{th} fold as testing, the first $k-1$ as training.

figure from ML for Predictive Data Analytics textbook.

K-Fold Cross Validation (II)

Hence, we will have **k performance measures**

The *final* measure can be aggregation of k measures.

IMPORTANT

You do cross validation to find the best model.

After estimating the performance using k-fold cross validation, you train a new model (for deployment) on all data.

K-Fold Cross Validation Example

5-fold cross validation performed on the chest X-ray classification dataset.

Fold	Confusion Matrix				Classification Accuracy
			Prediction		
			lateral	*frontal*	
1	Target	*lateral*	43	9	81%
		frontal	10	38	
			Prediction		
			lateral	*frontal*	
2	Target	*lateral*	46	9	88%
		frontal	3	42	
			Prediction		
			lateral	*frontal*	
3	Target	*lateral*	51	10	82%
		frontal	8	31	
			Prediction		
			lateral	*frontal*	
4	Target	*lateral*	51	8	85%
		frontal	7	34	
			Prediction		
			lateral	*frontal*	
5	Target	*lateral*	46	9	84%
		frontal	7	38	
			Prediction		
			lateral	*frontal*	
Overall	Target	*lateral*	237	45	84%
		frontal	35	183	

Leave-One-Out Cross Validation (jackknifing)

Extreme form of k-fold cross validation when **k equals number of datapoints**.

Useful when you have small amount of data.

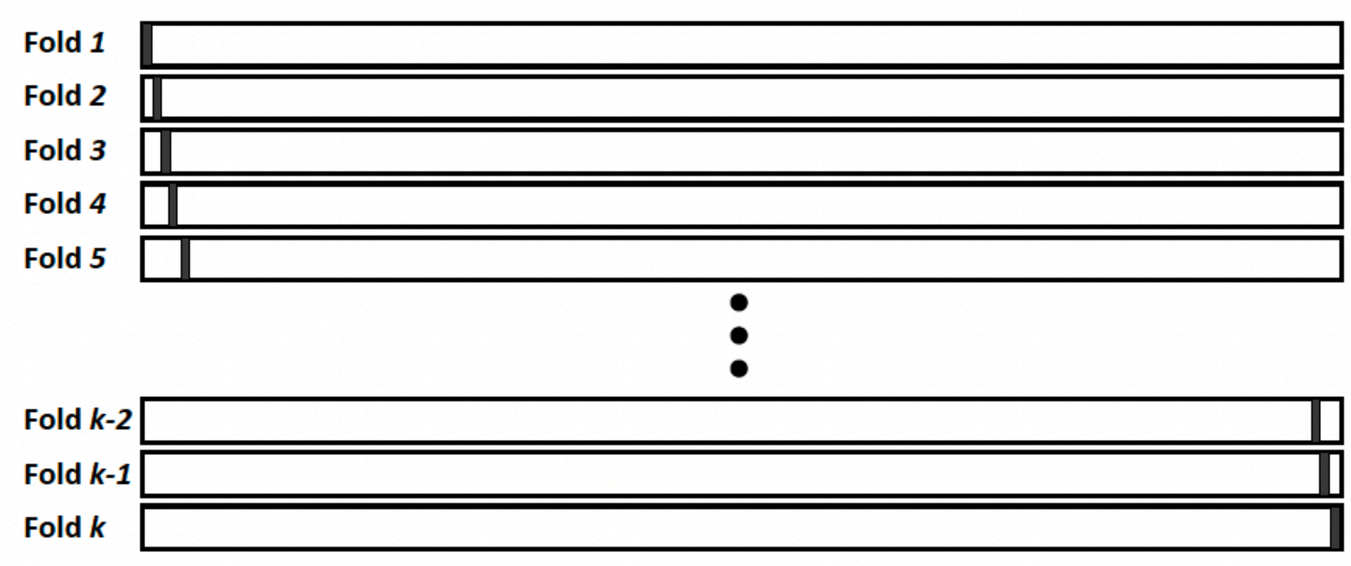


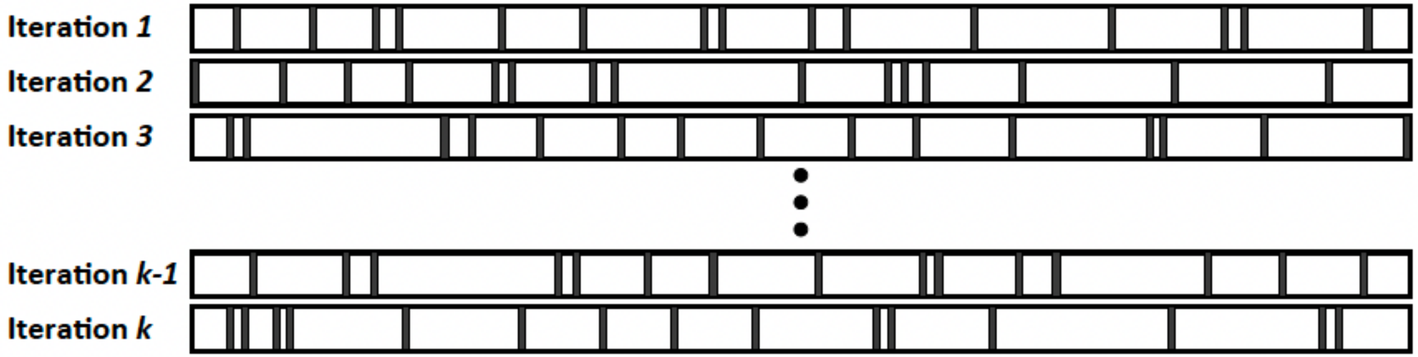
figure from ML for Predictive Data Analytics textbook.

What If You Have Very Small Dataset (e.g., 300 instances)

Bootstrapping

*Generation of training and testing sets involves a **random** selection of **m points** for testing, the rest is for training.*

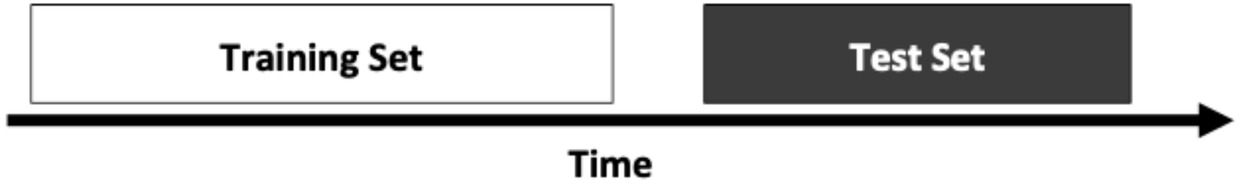
This is repeated many (**k**) times (e.g., 200 times)



What If Our Data is "Spread" over Time?

Out-of-Time Sampling

Test on a future chunk of data.



Important to take into consideration the notion of time (data examples are not iid anymore).

What can go wrong? Better to know the nature of your data (see next slide).

Food for Thought Example

Goal: Evaluate a model built to estimate the *daily energy demand* in a residential building based on features describing the family that lives in the house, the *weather* on a given day, and *time of the year*.

Training Sample: from a period in the summer.

Testing Sample: from a period in the winter.

The results of any evaluation would not provide a reliable measure of how likely the model might actually perform when deployed.

Make Sure Your Test Data is Good Representative of your Train

Also make sure your test data is a good representative of the data you will get after deployment.

Ex. Choose the periods for out-of-time sampling that the time spans are large enough to take into account any cyclical behavioural patterns.

Lets Look Into Two Key Aspects of Evaluation Together

- The **test set** to evaluate your model on.

- The **performance metric** used evaluate your model with.

Simplest Measure to Evaluate Your Model

Misclassification Rate

$$\text{misclassification rate} = \frac{\text{number of incorrect predictions}}{\text{total predictions}}$$

Confusion Matrix

		predictions	
		positive	negative
target	positive	**TP**	**FN**
	negative	**FP**	**TN**

Confusion Matrix Explained (I)

		predictions	
		positive	negative
target	positive	TP	FN
	negative	FP	TN

True Positive (TP)

Data point from test set, with positive target, is predicated to have a positive target.

True Negative (TN)

Data point from test set, with negative target, is predicated to have a negative target.

Confusion Matrix Explained (II)

		predictions	
		positive	negative
target	positive	**TP**	**FN**
	negative	**FP**	**TN**

False Positive (FP)

Data point from test set, with negative target, is predicated to have a positive target.

False Negative (FN)

Data point from test set, with positive target, is predicated to have a negative target.

Confusion Matrix Explained (III)

		predictions	
		positive	negative
target	positive	TP	FN
	negative	FP	TN

Hits

Misses

Confusion Matrix and Statistical Hypothesis Testing

		predictions	
		positive	negative
target	positive	TP	FN
	negative	FP	TN

Type I error

Type II error

Null hypothesis: the person is innocent.

Type I Error

Incorrect rejection of null hypothesis.

Ex. Convicting an innocent person.

Type II Error

Fail to reject of null hypothesis.

Ex. Fails to convict someone who is guilty.

This slide is extra and you WONT be evaluated on!

Which *False* to Minimize?

		predictions	
		positive	negative
target	positive	TP	FN
	negative	FP	TN

Depending on application one may be more important than the other to minimize.

Null hypothesis: The person is healthy.

False Positive: The model predicts the person to have cancer, but the person does not have cancer.

False Negative: The model predicts the person is healthy, but the person has cancer.

Misclassification Rate in Terms of TP, TN, FP, and FN

Misclassification Rate

$$\text{misclassification rate} = \frac{\text{FP} + \text{FN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Classification Accuracy

$$\text{classification accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

More Performance Measures for Categorical Targets (I)

Highlight different aspects of the performance.

True Positive Rate

Out of the positive targets, how many are classified correctly.

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

		predictions	
		positive	negative
target	positive	TP	FN
	negative	FP	TN

True Negative Rate (a.k.a Specificity)

Out of the negative targets, how many are classified correctly.

$$\text{TNR} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

		predictions	
		positive	negative
target	positive	TP	FN
	negative	FP	TN

More Performance Measures for Categorical Targets (II)

False Positive Rate

Out of the negative targets, how many are classified incorrectly.

$$\text{FPR} = \frac{\text{FP}}{\text{TN} + \text{FP}}$$

		predictions	
		positive	negative
target	positive	**TP**	**FN**
	negative	**FP**	**TN**

False Negative Rate

Out of the positive targets, how many are classified incorrectly.

$$\text{FNR} = \frac{\text{FN}}{\text{TP} + \text{FN}}$$

		predictions	
		positive	negative
target	positive	**TP**	**FN**
	negative	**FP**	**TN**

Precision and Recall

Precision

Captures how often, when a model makes a positive prediction, the prediction turns out to be correct.

$$\frac{\text{TP}}{\text{TP} + \text{FP}}$$

		predictions	
		positive	negative
target	positive	TP	FN
	negative	FP	TN

Recall (a.k.a TPR, a.k.a Sensitivity)

How confident we can be that all the instances with the positive target level have been found by the model.

$$\frac{\text{TP}}{\text{TP} + \text{FN}}$$

		predictions	
		positive	negative
target	positive	TP	FN
	negative	FP	TN

F_1 Measure (other names: F Measure, F Score)

It computes the harmonic mean of *Precision* and *Recall.*

$$F_1 \text{ measure} = 2 \times \frac{(\text{precision} \times \text{recall})}{(\text{precision} + \text{recall})}$$

Harmonic mean focuses on smaller values;

Hence, F_1 measure highlights shortcomings in our models rather than hide.

In-Class Example

Given the following sample test set, compute each of the metrics introduced in previous slides.

ID	Target	Pred.	Outcome
1	spam	ham	FN
2	spam	ham	FN
3	ham	ham	TN
4	spam	spam	TP
5	ham	ham	TN
6	spam	spam	TP
7	ham	ham	TN
8	spam	spam	TP
9	spam	spam	TP
10	spam	spam	TP

ID	Target	Pred.	Outcome
11	ham	ham	TN
12	spam	ham	FN
13	ham	ham	TN
14	ham	ham	TN
15	ham	ham	TN
16	ham	ham	TN
17	ham	spam	FP
18	spam	spam	TP
19	ham	ham	TN
20	ham	spam	FP

Misclassification Rate

ID	Target	Pred.	Outcome
1	spam	ham	FN
2	spam	ham	FN
3	ham	ham	TN
4	spam	spam	TP
5	ham	ham	TN
6	spam	spam	TP
7	ham	ham	TN
8	spam	spam	TP
9	spam	spam	TP
10	spam	spam	TP

ID	Target	Pred.	Outcome
11	ham	ham	TN
12	spam	ham	FN
13	ham	ham	TN
14	ham	ham	TN
15	ham	ham	TN
16	ham	ham	TN
17	ham	spam	FP
18	spam	spam	TP
19	ham	ham	TN
20	ham	spam	FP

We have 20 examples, and out of which, 5 are incorrect. Hence, misclassification rate is $\frac{5}{20} = 0.25$.

Classification rate is 0.75.

Choose A Positive Level

Typically, the level that is of most interest is referred to as the positive level.

For this example (email classification), identifying spam emails is the most important issue. Hence, choose spam as the positive level.

Other examples (disease diagnosis), a confirmation that a patient has the disease would most likely be the positive level.

Confusion Matrix

ID	Target	Pred.	Outcome
1	spam	ham	FN
2	spam	ham	FN
3	ham	ham	TN
4	spam	spam	TP
5	ham	ham	TN
6	spam	spam	TP
7	ham	ham	TN
8	spam	spam	TP
9	spam	spam	TP
10	spam	spam	TP

ID	Target	Pred.	Outcome
11	ham	ham	TN
12	spam	ham	FN
13	ham	ham	TN
14	ham	ham	TN
15	ham	ham	TN
16	ham	ham	TN
17	ham	spam	FP
18	spam	spam	TP
19	ham	ham	TN
20	ham	spam	FP

		predictions	
		spam	ham
target	spam	6	3
	ham	2	9

Lets take a look here to know what kind of mistakes the model is making

What Kind of Mistakes the Prediction Model is Making?

		predictions	
		spam	ham
target	spam	6	3
	ham	2	9

The models makes a prediction of *ham incorrectly 3 times out of 9 times that the correct prediction should be spam.* —> *33.333%*

The model makes a prediction of *spam incorrectly just 2 times out of the 11 times that the correct prediction should be* ham. —> 18.182%

Insight: when the model makes a mistake, it more commonly incorrectly predicts the *spam* level as the *ham* level.

You probably want a model that its FN is low in this case! So that you won't miss spams.

Precision and Recall

		predictions	
		spam	ham
target	spam	6	3
	ham	2	9

Precision: How often emails marked as spam actually are spam.

$$\text{precision} = \frac{6}{(6+2)} = 0.750$$

Recall: How often the spam messages in the set were actually marked as *spam.*

$$\text{recall} = \frac{6}{(6+3)} = 0.667$$

More Interpretation

		predictions	
		spam	ham
target	spam	6	3
	ham	2	9

How likely it is that a genuine *ham* email could be marked as *spam*

$$1 - \text{precision} = 25\,\%$$

How likely it is that a spam email will be missed by the system and end up in our inbox.

$$1 - \text{recall} = 33.333\,\%$$

F_1 Measure

$$F_1\text{measure} = 2 \times \frac{\left(\frac{6}{(6+2)} \times \frac{6}{(6+3)}\right)}{\left(\frac{6}{(6+2)} + \frac{6}{(6+3)}\right)} = 0.706$$

These measures, we looked into from confusion matrix, are particularly important when one target outcome is much more important that the other.

e.g., A prediction that a patient has a disease is much more important than a prediction that a patient does not.

What If All Target Levels Are of Equal Important? (use average class accuracy).

Average Class Accuracy (ACA)

$$\text{average class accuracy} = \frac{1}{|\,\text{levels}(t)\,|} \sum_{l \in \text{levels}(t)} \text{recall}_l$$

where levels(t) is the set of levels that the target feature, t, can assume.

This is using *arithmetic mean; susceptible to influence of large outliers.*

Hence, do ACA using harmonic mean

$$\text{average class accuracy}_{HM} = \frac{1}{\frac{1}{|\,\text{levels}(t)\,|} \sum_{l \in \text{levels}(t)} \frac{1}{\text{recall}_l}}$$

Compute Classification Accuracy in an Example for Two Models

Train models on a churn prediction problem.

KNN

		predictions	
		not-churn	*churn*
target	*not-churn*	90	0
	churn	9	1

$$\text{acc} = 91\,\%$$

Naive Bayes model

		predictions	
		not-churn	*churn*
target	*not-churn*	70	20
	churn	2	8

$$\text{acc} = 78\,\%$$

Is KNN a better model, then? (see next slide)

Compute ACA

KNN

		predictions	
		not-churn	*churn*
target	*not-churn*	90	0
	churn	9	1

$$\text{ACA} = \frac{1}{2}(1 + 0.1) = 55\,\%$$

Naive Bayes model

		predictions	
		not-churn	*churn*
target	*not-churn*	70	20
	churn	2	8

$$\text{ACA} = \frac{1}{2}(0.778 + 0.8) = 78.889\,\%$$

Naive Bayes is better. When there is target level imbalance, rely on ACA instead of accuracy (ACC).

Compute ACA_{HM}

KNN

		predictions	
		not-churn	*churn*
target	*not-churn*	90	0
	churn	9	1

$$\text{ACA}_{HM} = \frac{1}{\frac{1}{2}(\frac{1}{1.0} + \frac{1}{0.1})} = 18.2\,\%$$

Naive Bayes model

		predictions	
		not-churn	*churn*
target	*not-churn*	70	20
	churn	2	8

$$\text{ACA}_{HM} = \frac{1}{\frac{1}{2}(\frac{1}{0.778} + \frac{1}{0.8})} = 78.873\,\%$$

The harmonic mean results in a more pessimistic view of model performance than an arithmetic mean.

Measuring Profit and Loss

What is the cost of doing a mistake or predicting correctly, ...?

Consider a scenario where different classes have different importance to you.

We can calculate *profit* and *loss* that arise from each prediction.

A profit matrix

		predictions	
		positive	negative
target	positive	$\mathbf{TP_{profit}}$	$\mathbf{FN_{profit}}$
	negative	$\mathbf{FP_{profit}}$	$\mathbf{TN_{profit}}$

Multiply values from the confusion matrix with the corresponding value in the profit matrix.

Compute Profit of the Models in the Churn Example

Note: it's not always possible/feasible to construct profit matrix.

Profit matrix

		predictions	
		not-churn	*churn*
target	*not-churn*	140	-100
	churn	-700	100

KNN

		predictions	
		not-churn	*churn*
target	*not-churn*	90	0
	churn	9	1

$$\textbf{profit} = 90 * 140 + 9 * (-700) + 100 = 6400$$

Naive Bayes model

		predictions	
		not-churn	*churn*
target	*not-churn*	70	20
	churn	2	8

$$\textbf{profit} = 70 * 140 + 2 * (-700) + 20 * (-100) + 8 * 100 = 7200$$

Takeaways

- ***Concept of evaluation*** — *Standard approach, confusion matrix.*

- ***Evaluation experiment*** — *Sampling approaches, validation, bootstrapping.*

- ***Performance measure for categorical target*** — *precision, recall, F-measure, average class accuracy*